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1. Short Answer Questions (20 pt)

a. What is the difference between the Kiefer-Wolfowitz method and the Random Directions method? And what are the strengths of each?

b. Give an example of a function were Newton-Raphson will likely fail to converge. Explain why.

1. Short Answer Questions

2. Ratio of Normals (20 pt)

Let $X, Y \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$. Show that

$$Z = \frac{X-Y}{X+Y} \sim \text{Cauchy}.$$

Hint: Try change of variables with $W = X + Y$, and recall that the normal pdf and Cauchy pdfs are[1]

$$f_{\text{normal}}(t) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{t^2}{2}\right) \text{ and } f_{\text{cauchy}}(t) = \frac{1}{\pi}\frac{1}{1+t^2}.$$

[1]Second Hint: $\int_{-\infty}^{\infty} \frac{|t|}{\sqrt{2\pi}} \mathrm{e}^{-t^2/2} dt = \sqrt{\frac{2}{\pi}}$

2. Ratio of Normals

3. Importance Sampling (20 pt)

We want to estimate the integral[2]

$$\int_0^1 \mathrm{e}^x \sin(\pi x) dx.$$

Which of the following sampling distributions on $[0, 1]$ results in the smallest variance?[3]

$$f_1(x) = \frac{2}{\mathrm{e}^2 - 1}\mathrm{e}^{2x} \quad \text{or} \quad f_2(x) = 2\sin^2(\pi x)$$

[2]Fun Fact: the integral equals $\frac{\pi}{1+\pi^2}(\mathrm{e}+1)$.

[3]Hint: If Z has pdf $f(z)$ and $g(z)$ is some function, then $\mathrm{Var}\,(g(Z)/f(Z)) = \mathrm{E}[g(Z)^2/f(Z)^2] - \mu^2$ where μ is the mean.

3. Importance Sampling

4. Statification (20 pt)

We want to use Monte Carlo to estimate the integral of

$$f(x) = \begin{cases} -x & x \in [-2, 0] \\ x & x \in [0, 1] \end{cases}$$

on the interval $[-2, 1]$ using the stratified estimator

$$\hat{\mu} = \frac{2}{m} \sum_{i=1}^{m} (-X_i) + \frac{1}{n-m} \sum_{j=1}^{n-m} Y_j$$

where n is fixed, $m \in [1, n-1]$, $X_i \overset{\text{iid}}{\sim} \text{Uniform}\,[-2, 0]$, and $Y_i \overset{\text{iid}}{\sim} \text{Uniform}\,[0, 1]$. Find the value of m that minimizes the variance. That is $m = \delta n$ for some $\delta \in (0, 1)$, which may not be an integer.

4. Statification

5. At a loss (20 pt)

Consider the function, $g(t) = \sqrt{1+t^2} - 1$. Show that $g(t)$ is convex with minimum at $g(0) = 0$.

Then for data (x_i, y_i), find the gradient of

$$L(\theta) = \sum_{i=1}^{n} \left[\sqrt{1 + \left(y_i - \sum_{j=1}^{k} \theta_j x_{i,j} \right)^2} - 1 \right]$$

5. At a loss

6. More Acceptance-Rejection (20 pt)

Let $f(t) = \mathrm{e}^{-t}/(1+\mathrm{e}^{-t})^2$ be the logistic distribution and $h(t) = \mathrm{e}^{-|t|}/2$ be the Laplace distribution. If we want to use acceptance-rejection sampling to use h to sample from f, find c and g in the decomposition $f(t) = cg(t)h(t)$. What is the probability of acceptance?

6. More Acceptance-Rejection

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